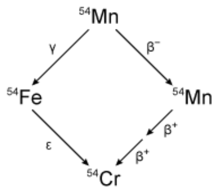
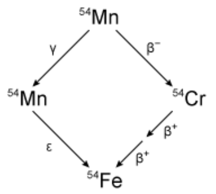
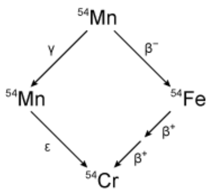
Manganese-54 can undergo a β⁻ decay that is followed by two positron emissions (β⁺ decays), or it can undergo a gamma decay (γ) that is followed by electron capture ε (a form of β⁺ decay). Which of the following diagrams correctly describes the nuclear decay chain of ^{54}Mn?

A.


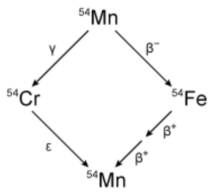
[3%]

B.


[12%]

C.


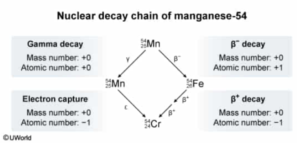
[80%]

D.


[4%]

Correct
79% answered correctly

Explanation:



Radioactivity is the spontaneous emission of radiation (photons or particles) from an unstable atomic nucleus. Although several types of nuclear radiation exist, the three most common types are alpha decay, beta decay, and gamma decay.

During gamma decay, an unstable nucleus in an **excited state** releases excess energy by emitting a gamma ray (γ), which is a **high-energy photon** (an electromagnetic wave packet). One characteristic of gamma emission is that the number of protons and neutrons in the nucleus remains the same. As a result, the **elemental identity** of a nucleus is **unchanged** by gamma emission. For example, the gamma decay of ^{54}Mn yields:

$$^{54}_{25}\text{Mn} \rightarrow {}^{54}_{25}\text{Mn} + {}^{0}_{0}\gamma$$

Beta decay can occur in three forms: β⁻ decay (electron emission), β⁺ decay (positron emission), and electron capture. In all three forms of beta decay, the mass number remains **unchanged** but the atomic number either **increases** (β⁻ decay) or **decreases** (β⁺ decay, electron capture) by **1 unit**.

In **β⁻ decay**, a neutron converts to a nuclear proton and emits an electron (e^-) and an antineutrino ($\overline{\nu}_e$). For example, the β⁻ decay of ^{54}Mn yields:

$$^{54}_{25}\text{Mn} \rightarrow {}^{54}_{26}\text{Fe} + e^- + \overline{\nu}$$

In **β⁺ decay**, a nuclear proton converts to a neutron (the opposite of β⁻ decay) and ejects a positron (e^+) and a neutrino (ν_e). For example, the β⁺ decay of ^{54}Fe yields:

$$^{54}_{26}\text{Fe} \rightarrow {}^{54}_{25}\text{Mn} + e^+ + \nu_e$$

And a subsequent β⁺ decay of ^{54}Mn yields:

$$^{54}_{25}\text{Mn} \rightarrow {}^{54}_{24}\text{Cr} + e^+ + \nu_e$$

In **electron capture**, a proton captures an electron near the nucleus and converts to a neutron without a positron or electron emission. For example, nuclear decay of ^{54}Mn by electron capture yields:

$$^{54}_{25}\text{Mn} + e^- \rightarrow {}^{54}_{24}\text{Cr}$$

Based on these principles and the examples given above, only the diagram with ^{54}Fe as the product of β⁻ decay and ^{54}Cr as the electron-capture product **(Choice C)** correctly describes the full decay chain of ^{54}Mn.

Educational objective:
In gamma decay, a high-energy photon is released from a nucleus and the mass number, atomic number, and elemental identity remain unchanged. In all forms of beta decay, the mass number of an atomic nucleus remains unchanged but the atomic number increases (β⁻ decay) or decreases (β⁺ decay, electron capture) by 1 unit.

Time spent:0 sec
Subject:
General Chemistry

QID:402672
System:
Atomic Theory and Chemical Composition